

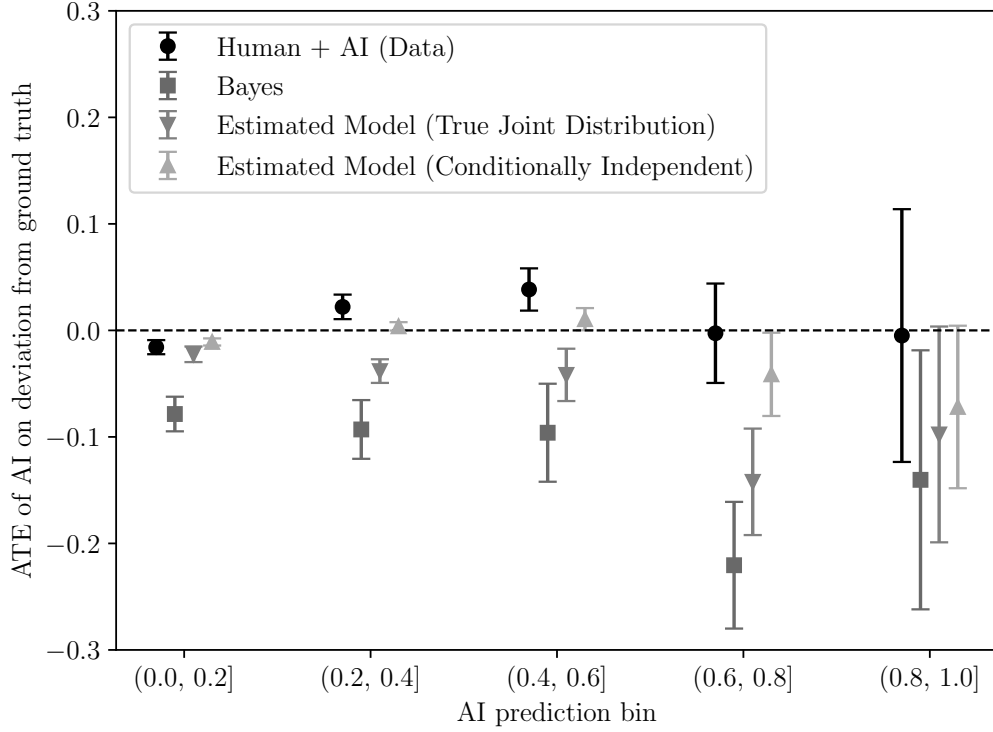
Table 3: Model Selection: Top Level with AI – IVGMM

	(1)	(2)	(3)	(4)	(5)	(6)
AUTOMATION BIAS (b)	0.29 (0.02)	0.34 (0.03)	0.35 (0.03)	0.19 (0.02)	0.27 (0.02)	0.35 (0.03)
OWN INFO BIAS (d)	1.10 (0.01)	1.08 (0.01)	1.05 (0.01)	1.07 (0.01)	1.06 (0.01)	1.05 (0.01)
CONSTANT	0.40 (0.03)	0.40 (0.04)	0.37 (0.04)	0.31 (0.03)	0.33 (0.03)	0.36 (0.04)
FOCAL s_A	✓	✓	✓	✓	✓	✓
OTHER s_A			✓			✓
FOCAL s_E		✓	✓		✓	✓
OTHER s_E			✓			✓
CLINICAL HISTORY s_E				✓	✓	✓
J-STATISTIC	18.23	16.86	0.17	4.3	10.7	0.09
MMSC-BIC	-23.96	-21.12	-8.27	-12.57	-10.39	-8.35
NUMBER OF MOMENTS	13	12	5	7	8	5
Observations	11420	11420	11420	11420	11420	11420
R-SQUARED	0.50	0.49	0.52	0.46	0.48	0.52

Note: This table summarizes estimates of b and d for different specifications of the update term. The models differ by whether the update term conditions on the signal s_E of the pathology at hand, the AI's and the radiologist's signal of other pathologies as well as the information that is provided in the clinical history of a patient. Each model is estimated via GMM. The reported MMSC-BIC statistic adjusts the J-statistic for the number of included parameters, awarding bonus terms for models with fewer parameters (see [Andrews and Lu \(2001\)](#) for details). This table presents a subset of the models that we select from. The full set of models included in the selection procedure are presented in table [C.20](#). The update term is estimated via random forest as described in appendix section [C.5.2](#). Standard errors are clustered at the radiologist level.

These deviations also explain the heterogeneous conditional average treatment effects documented in section [4](#). Figure [6](#) shows the treatment effect of AI on the deviation from ground truth of our radiologists along with three different models under which we compute the same treatment effects: the Bayesian benchmark, the model in equation [\(5\)](#) where radiologists use the correct updating term, and the selected model from table [3](#). As expected, the Bayesian would benefit significantly from gaining access to the AI signal. In fact, as indicated by the large reductions in the deviation from the ground truth, there is significant potential value in combining the human and the AI signal. Imposing the model in equation [\(5\)](#) with the correct updating term – column (6) – reduces these improvements and moves the implied treatment effects closer to the data. However, we see that throughout the entire signal range of s_A , such a decision-maker would still unambiguously benefit from gaining access to the AI. Only when we impose the selected update term, under which radiologists do not account for the dependence of signals, does the model generate the observed negative treatment effect

Figure 6: Model Treatment Effects



Note: This graph shows the observed conditional treatment effects of providing radiologists access to AI (Human+AI) and compares those to three different model-implied treatment effects: giving AI access to a Bayesian decision maker, giving AI access to a decision maker who acts according to the empirical version of equation 5, both under the correct updating term and when the decision maker treats the AI signal as conditionally independent.

conditional on intermediate AI predictions as in our treatment effect analysis. We see that such a model closes even more of the gap between the Bayesian treatment effects and the observed treatment effects.

It is significant that the model presented above is able to replicate the pattern of conditional treatment effects even though it is restrictive – it imposes linearity in log-odds and does not allow for heterogeneity in b and d across radiologists. Appendix C.5.5 investigates the linearity restriction in equation (7). We estimate a boosted tree that predicts the dependent variable in the equation using the same two independent variables in the equation. We find that the empirical analog to the relevant boundary of the decision-regions depicted in figure 5 is well-approximated using a linear model. Appendix C.5.6 investigates heterogeneity – it presents estimates that allow for b and d to vary with radiologists. We find that the estimated distributions of b_r and d_r are centered close to the point estimates above, with most of the estimated distributions of b_r and d_r in the range $[0.1, 0.4]$ and $[1.0, 1.2]$ respectively.

Thus, consistent with the conclusions above, we find evidence for automation neglect, but no meaningful own-information bias or neglect. These results suggest that the estimated model represents a good approximation to the experimental data.

Taken together, the results indicate that while there are large potential gains from combining radiologists’ assessments with AI predictions, biases in radiologists’ use of AI assistance undercuts these gains. We find that radiologists exhibit both automation-neglect and signal dependence neglect. These mistakes lead to no better diagnostic performance when provided with AI assistance. One approach for addressing this issue is to provide radiologists with better training, an avenue we leave for future research. A second approach is to design better collaborative systems that are built on predicting the types of cases in which human experts outperform AI predictions and vice-versa, and the types of cases in which AI-assisted humans do best, an issue we turn to next.

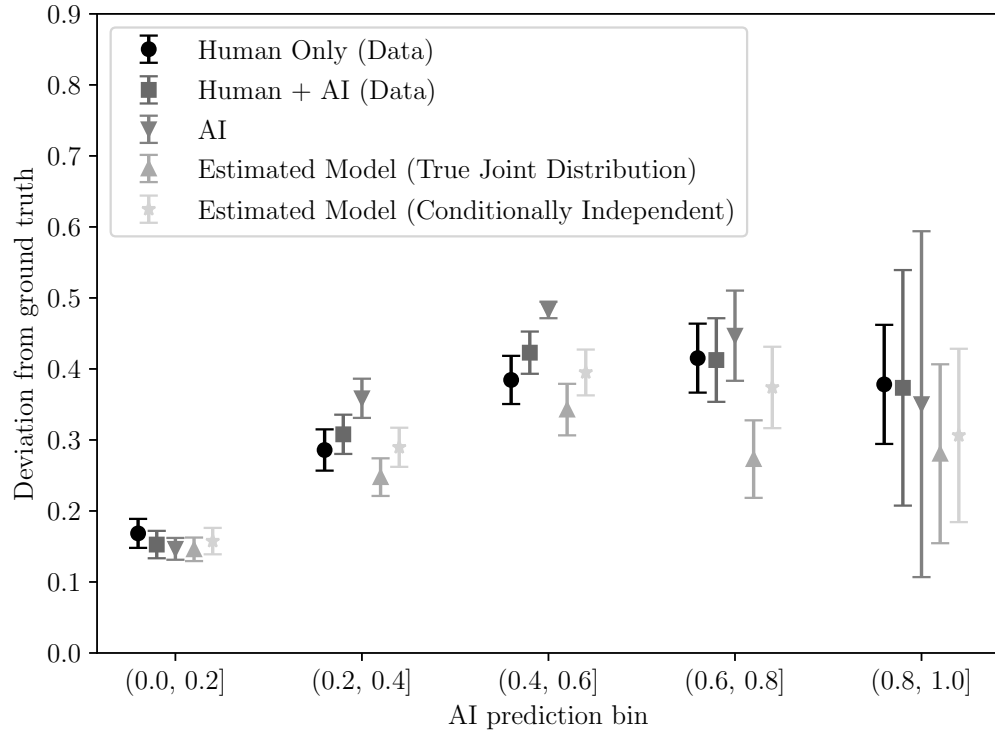
6 Designing Human-AI Collaboration

We now consider the design of collaborative systems between AI and radiologists. The designs we consider restrict attention to systems that use the AI signal to delegate a case to one of three modalities – radiologists alone, the AI alone, or the radiologist with access to the AI. As a warm-up for this exercise, it is useful to examine the predictive performance of the different modalities, conditional on s^A (see figure 7). Recall from the conditional treatment effect analysis, radiologists’ assessments improve with AI in the lowest and the two highest bins of AI signals. Figure 7 also shows that in two out of three ranges in which the AI improves radiologists’ decision-making that its own performance is even better than the performance of the radiologists with AI.

However, this figure misses the differences in the human time costs across modalities. Our analysis of the estimated treatment effects shows that radiologists take more time when provided with AI predictions. Moreover, using AI predictions alone saves on costly human effort. This points to the conclusion that in most cases where AI improves decision-making, one is at least as well off by relying exclusively on AI prediction. But using only AI or never assisting humans with AI predictions may come at the cost of performance in cases when using the human alone is superior.

Motivated by these observations, we now examine if there is a trade-off between the marginal costs of human effort and diagnostic performance.

Figure 7: Model Deviation from Ground Truth



Note: This figure shows the performance of the different modalities that we consider for the optimal collaborative system. Cases are either decided by only the radiologist, only the AI, or the radiologist with access to the AI. The performance measures for Human Only and Human + AI are constructed from our treatment effect analysis.